MIL-S-60891B(MU)
18 July 1972
SUPERSEDING
MIL-S-60891A(MU)
5 March 1969

MILITARY SPECIFICATION

SIGHT, REFLEX: M73

1. SCOPE

1.1 This specification establishes the requirements for manufacture and acceptance of the Sight, Reflex: M73. It will be referred to herein as a sight. Primarily for use in helicopters (see 6.1), this sight incorporates a mechanism to elevate and depress its line of sight which is fixed in azimuth.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-T-60	Tape, Pressure-Sensitive, Adhesive, Waterproof, For Packaging

Military

MIL-O-13830	Optical Components of Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-13988	Paper, Lens, Tissue, Antitarnish; Wrapping
MIL-P-14232	Parts, Equipment and Tools for Army Materiel: Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition Maintenance and Disposition of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

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DRAWINGS

U. S. Army, Frankford Arsenal

F8565556	Pressure Tester - Two Station
A8658940	Inspection Aid (Gunner's Quadrant)
F10553252	Sight, Reflex: M73

PACKAGING DATA SHEET

10553252	Packaging of Sight, Reflex: M73

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.2 Fabrication.- The sight shall be fabricated in accordance with Drawing F10553252 and drawings pertaining thereto.

3.3 General specifications.- The contractor shall be responsible for adherence to and compliance with the following requirements of MIL-O-13830 and MIL-F-13926.

3.3.1 MIL-O-13830 requirements:

(a) Lenses

(b) Prisms

(c) Cleanliness

3.3.2 MIL-F-13926 requirements:

(a) Order of precedence

(b) Dimensions and tolerances

(c) Inorganic protective surface finishes

(d) Part identification and marking

(e) Workmanship

3.4 Environmental performance.

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3.4.1 Storage at extreme temperatures.- The sight shall exhibit no evidence of damage after storage at -65° Fahrenheit (F) and at +160°F.

3.4.2 Operability at extreme temperatures.- While the sight is thermally stabilized at -40°F and at +150°F, the elevation/depression knob running torque shall be four (4) to 12 pound inches. The elevation/depression knob, light intensity control knob, polarizing filter lever, and lamp housing assemblies shall be manually operable throughout their respective operational range.

3.4.3 Humidity.- Exclusive of the Filter (Riveted) 10553390; Plug, Straight 10553320; Plug MS3116-F-8-3P; and Connector MS3112E-10-6P, the sight shall exhibit no damage after exposure to a 97% relative humidity for 24 hours at +160°F, followed by four (4) hours at between +60°F and +90°F with condensation taking place on the sight.

3.4.4 General vibration requirements.- The projected reticle image of the sight shall not flicker during vibration; the sight shall demonstrate the line of sight retention specified in each vibration requirement; and the sight shall evidence no physical damage or electrical circuit malfunction during or subsequent to meeting each vibration requirement between +60°F and +90°F.

3.4.4.1 Vibration A.- The line of sight shall not change more than one mil after a three hour vibration, in each plane of three mutually perpendicular planes, at the sinusoidal frequencies and amplitudes depicted in Figure 1. During each first hour period, the frequency shall vary from 5 to 400 to 5 cycles-per-second (cps) at a logarithmic rate over a 15 ± 1.0 minute period. A thirty minute dwell shall occur at each of the most significant resonant frequencies (not exceeding four) exhibited by the sight during the first hour of each three hour vibration in each plane.

3.4.4.2 Vibration B.- The line of sight shall not change more than two mils during the two following vibration phases:

a. PHASE I.- Vibration in a vertical and lateral plane for 5 ± 0.5 minute at the frequencies and amplitudes in Table I.

TABLE I - PHASE I VIBRATION

FREQUENCY (CPS) (BOTH PLANES)	DOUBLE AMPLITUDE (IN INCHES) (VERTICAL PLANE)	(LATERAL PLANE)
5.4	0.128	0.066
11.9	0.038	0.032
47.5	0.008	0.006
65.0	0.006	0.004

b. PHASE II.- Vibration in a vertical and lateral plane varied continuously over a frequency range from 5 to 65 cps and return to 5 cps at a double amplitude of 0.072 inch or an acceleration of two G(s) whichever is lesser (see Figure 1) for a 5 ± 1.0 minute period in each plane.

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3.5 General performance (between +60°F and +90°F).

3.5.1 Inclinometer accuracy.- The inclinometer ball shall center within 0.03 inch.

3.5.2 Manual operability.- The elevation/depression knob, light intensity control knob, polarizing filter lever, and lamp housing assemblies, shall exhibit ease of manipulation while performing their intended function.

3.5.3 Torque.- The elevation/depression knob shall exhibit a running torque of between four (4) and seven (7) pound inches.

3.5.4 Illumination.- Light intensity shall attenuate with counterclockwise (ccw) rotation of the light intensity control. Clockwise (cw) rotation of this control shall increase light intensity.

3.5.5 Projected reticle pattern definition.- The projected reticle pattern image lines shall show no breaks and be clearly defined when viewed from a normal operating position in a darkened area and against a bright sky.

3.5.6 Sealing.- The sight shall show no evidence of leakage in excess of 0.2 pound per square inch when subjected to an internal pressure of at least five pounds per square inch, gage pressure, (psig) for a minimum of two hours.

3.5.7 Interior atmosphere.- The interior atmosphere of the sight shall contain dry nitrogen having a dew point at least as low as minus 25°F with an internal pressure of 0.10 to 0.25 psig.

3.6 Optical performance (between +60°F and +90°F). The following optical requirements shall be met with the sight orientated in accordance with Figure 2 and the following:

(a) Surface "X" shall be plumb within one mil.

(b) Surface "Y" shall incline toward an observer 6° ± 0.5' with respect to a plumb line.

(c) Elevation control scale shall indicate zero.

(d) The nominal dc electrical potential shall be 28±1.0 volt.

3.6.1 Optical line of sight.- The line of sight, through the central dot of the projected reticle pattern image, shall be parallel to a horizontal plane within 1.5 mils and parallel to surface "X" within 1.5 mils.

3.6.2 Image tilt.- The image of the vertical reticle line shall be parallel to a plumb line within one degree.

3.6.3 Elevation and depression accuracy.- Clockwise rotation of the elevation knob shall depress the line of sight and position it within

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three (3) mils of indicated elevation knob scale settings from zero to +125 mils. Counterclockwise rotation of the elevation knob shall elevate the line of sight and position it within three (3) mils of indicated elevation knob scale settings from zero to -125 mils.

3.6.4 Plumb travel.- With the line of sight in coincidence with a plumb target line at zero elevation, the optical line of sight shall not deviate in azimuth more than one mil while being elevated and depressed 100 mils.

3.6.5 Reticle pattern accuracy.- The total length of the vertical and horizontal reticle lines shall each be 100 ±3.0 mils.

3.6.6 Parallax.- When viewing targets between 500 meters and infinity, parallax at the center of the reticle image shall not exceed 0.5 mil over a centered 1.75 inch minimum projected aperture on the beamsplitter.

3.7 Interchangeability.

3.7.1 Components.- The sight shall meet the requirements of 3.3 to 3.6 without the need for selective components. Compatibility shall exist between mechanical interfaces of components required for sight maintainability.

3.7.2 Assemblies.- Compatibility shall exist between the mechanical interfaces and between the optical, mechanical and electrical performance characteristics of any random combination of head assembly and body assembly.

3.8 Reliability.- The reliability of the sight shall be 95 percent at a 90 percent confidence level.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-O-13830 and MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) M73 sights plus three (3) each of all items covered by SQAP for the sight.

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The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAPs, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables II and III herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE II - CLASSIFICATION OF DEFECTS

CLASS	REQUIREMENT	TEST PROCEDURES
CRITICAL: NONE		
MAJOR: AQL 1.-% Defective		
101. Vibration (General)	3.4.4	4.6.6
101a. Vibration B (Phase I)	3.4.4.2a	4.6.6.2
101b. Vibration B (Phase II)	3.4.4.2b	4.6.6.2
102. Inclinometer accuracy	3.5.1	4.6.7.1
103. Manual operability	3.5.2	4.6.7.2
104. Torque	3.5.3	4.6.7.3

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105.	Illumination	3.5.4	4.6.7.4
106.	Projected reticle pattern definition	3.5.5	4.6.7.5
107.	Optical line of sight	3.6.1	4.6.7.6
108.	Image tilt	3.6.2	4.6.7.7
109.	Elevation and depression accuracy	3.6.3	4.6.7.8
110.	Plumb travel	3.6.4	4.6.7.9
111.	Reticle pattern accuracy	3.6.5	4.6.7.10
112.	Parallax	3.6.6	4.6.7.11
113.	Cleanliness	3.3.1(c)	4.6.7.12
114.	Sealing	3.5.6	4.6.3
115.	Interior atmosphere	3.5.7	4.6.5

MINOR: NONE

NOTE: The tests in Table II shall be conducted under the standard ambient test condition requirements in the "Test Facilities" requirement of MIL-F-13926.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Control tests.

4.4.1 Special sampling.- Samples of completely assembled sights shall be selected from lots at random using inspection level S-3 Table I of MIL-STD-105. Acceptance or rejection of the lot shall be in accordance with AQL 2.5 of Table II-A of MIL-STD-105. Head and body assembly samples when assembled into sights shall meet the requirements and tests in Table II prior to and after meeting the requirements and tests in Table III.

TABLE III

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURES
301.	Storage at extreme temperature	3.4.1	4.6.1
302.	Operability at extreme temperatures	3.4.2	4.6.2
303.	Humidity	3.4.3	4.6.4
304.	Interchangeability (assemblies only)	3.7.2	4.6.8.2

4.4.1.1 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.4.2 First article sample.- Unless otherwise specified in the contract examinations listed herein shall be conducted on one unit of the initial production sample and shall not again be performed on the remaining production quantities, provided the manufacturing processes and techniques used to produce the samples have not been changed or revised.

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURE
305.	Interchangeability (components)	3.7.1	4.6.8.1
306.	Reliability	3.8	4.6.9
307.	Vibration "A"	3.4.4;3.4.4.1	4.6.6;4.6.6.1

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4.4.3 Certification.- Certification shall include test data and results for all characteristics specified in Table IV. Certification shall be provided prior to performing final acceptance inspection, and remain in effect throughout the contract for Government acceptance. This does not relieve the contractor from inspection throughout the contract production, including recording data and results therefrom. Recorded data and results shall be made available to the Government upon request. A new certification will be required should any changes occur in materials, manufacturing processes and techniques previously certified.

TABLE IV

CHARACTERISTIC	REQUIREMENT	TEST DATA TO COMPLY WITH
401. Materials	3.1	Applicable drawings
402. Fabrication	3.2	Standards,
403. General specification (MIL-F-13926)	3.3.2	Specifications,
404. General specification (MIL-0-13830)	3.3.1 (except 3.3.1(c))	and SQAPs
405. Interchangeability (components only)	3.7.1	

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the Technical Data Package (TDP).

4.5.2.2 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

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4.5.2.3 Inspection equipment design requirements.- Inspection equipment provided in accordance with 4.5.2.2 shall meet the following requirements:

a. Environmental and mechanical test equipment.- Temperature, humidity, and vibration test equipment shall conform to applicable requirements in the "Test Facilities" requirement of MIL-F-13926. Standard vibration equipment provided shall have provisions for controlling the frequency and amplitude of vibration in accordance with 3.4.4.1 and 3.4.4.2. Vibration equipment shall have provisions for holding securely the sight throughout the vibration tests, and allow it to be orientated in three mutually perpendicular planes for Vibration A and in a vertical and lateral plane for Vibration B. A standard time measuring instrument shall be provided to control vibration periods. Also provided shall be a standard torque wrench with appropriate adapters.

b. Optical test equipment.- Inspection equipment shall be provided to hold securely the sight under test while accurately positioning it along specified optical and geometric lines - as required in 3.6 and the test in 4.6.7. Equipment shall be capable of crossleveling the medium used to position the sight and to accurately provide the required elevation and depression limits. A device having at least the accuracy of the inspection aid depicted on Drawing A8658940 (see 4.5.2.4, Item 6) shall be provided to measure excursions in elevation and depression. A suggested test set-up is shown in Figure 3. A ten to fifteen power (10x to 15x) auxiliary telescope with a reticle indicating acceptance limits for all pertinent optical requirements shall also be provided.

c. Electrical test equipment.- An electrical test circuit external to the sight shall be provided. When used in conjunction with an electrical power source, it shall provide the various direct current (dc) potentials in proper polarity required by individual tests. It shall not be subject to the mechanical vibration of the vibration test, and it shall be connected electrically to items under tests by flexible conductors. This test circuit shall incorporate the required electrical instrumentation (see 4.5.2.4, Items 3 to 5 inclusive) to measure and monitor specified electrical characteristics of the items under test, and it shall provide a rheostat and switch for testing both filaments of the dual filament reticle lamp.

4.5.2.4 Test equipment accuracy.- In conjunction with 4.5.2.3, test equipment shall have at least the accuracy depicted for the following equipment:

ITEM	NOMENCLATURE	DESCRIPTION
1.	Pressure Tester - Two Station	Conforming to Drawing F8565556
2.	Dew-point Tester w/Regulator	Standard Commercial Quality
3.	Voltmeter	0-50 Vdc; full scale accuracy ±2%
4.	Ammeter	0-1.0 Amp dc; full scale accuracy ±2%
5.	Reticle light intensity control	Resistor, variable 100 ohms ±10%; 25 watt
6.	Inspection Aid (Gunner's Quadrant)	Conforming to Drawing A8658940

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4.6 Test methods and procedures.

4.6.1 Storage at extreme temperatures.- Utilizing test equipment conforming to 4.5, gradually change chamber temperature to prevent thermal shock until the external surface of the sight is thermally stabilized at each extreme temperature specified in 3.4.1. Maintain each extreme temperature for an uninterrupted period of at least 24 hours. Upon completion of each exposure the temperature shall be gradually returned to between +60°F and +90°F. Subsequent to completion of each temperature excursion, the sight shall be examined to determine that no evidence of damage exists.

4.6.2 Operability at extreme temperatures.- Utilizing test equipment conforming to 4.5, gradually change chamber temperature to prevent thermal shock and until external surface of the sight is thermally stabilized at temperatures in 3.4.2. Maintain extreme temperature for only the uninterrupted period of time required to perform the test and examinations specified herein. The test and examination in 4.6.7.2 shall be performed to verify manual operability of mechanical assemblies at each extreme temperature in compliance with 3.4.2. Utilize the same test procedure specified in 4.6.7.3 and determine that running torque of the elevation knob is within the limits specified in 3.4.2. Upon completion of all examinations at each extreme temperature, the temperature shall be gradually returned to between +60°F and +90°F.

4.6.3 Sealing.- Unless otherwise specified herein, this test shall be conducted in accordance with applicable requirements in the "Test Facilities" requirement of MIL-F-13926, utilizing test equipment conforming to Drawing F8565556 (see 4.5.2.4) and instructions provided thereon. The sight shall be internally pressurized with dry nitrogen gas to five pounds-per-square-inch gage pressure (psig) for not less than two hours. Throughout this time, no pressure drop greater than 0.2 psig shall be evident. Flush the sight with dry nitrogen gas through a regulator to prevent excessive internal pressure during flushing. Utilize a standard commercial type dew point tester and continue flushing until the gas bled from the sight exhibits a dew point at least as low as -25°F and an internal pressure of 0.10 to 0.25 psig. The sight shall then be sealed.

4.6.4 Humidity.- Utilize test equipment conforming to 4.5 and perform this test subsequent to the sealing test and prior to performing the interior atmosphere test. It shall be conducted in accordance with Procedure III of the humidity test in MIL-F-13926, except that the sight shall be exposed for at least 24 hours to a 97% relative humidity (RH) at an ambient temperature of +160°F. Chamber temperature shall then be reduced to between +60°F and +90°F with condensation and the sight exposed to this condition for at least four (4) hours, within which time chamber temperature may be adjusted to obtain other condensation periods. The humidity phase of this test shall now cease and the external surfaces of the sight shall be dried by wiping all accessible areas. Drying by air blast or in an oven shall not be permitted.

"CAUTION: Performance requirements of this specification require that specified items be protected from direct exposure to this humidity test. Such items, as are specified, shall first be wrapped with paper conforming to MIL-P-13988 and then completely covered with adhesive paper tape conforming to PP-T-60. Subsequent to this test, remove the protective cover-

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ing and verify that all previously covered surfaces are free of foreign matter and that the filter has suffered no color fading or clouding."

4.6.5 Interior atmosphere.- This test shall utilize the procedure in 4.6.3 except that the examination previously performed to detect excessive loss of pressure need not be repeated.

4.6.6 Vibration test requirements.- Perform vibration tests A and B utilizing test equipment conforming to 4.5 in accordance with the requirements and examinations specified herein. Secure the sight to the vibration table making all electrical connections in proper polarity, and apply a direct current (dc) potential of 30 volts (v) to the sight through its electrical input connector. Throughout vibration, electrical circuits shall be under constant surveillance. Monitor all electrical circuits to determine that no input current change occurs indicative of an intermittent electrical connection or short circuit. The electrical test switch shall be sequentially actuated to determine the absence of flicker in the dual filament reticle lamp in each switch position. Examine instrument lamps to determine that they remain securely mounted and exhibit no flicker. Prior and subsequent to vibration tests A and B, conduct test 4.6.7.1 to determine compliance with 3.5.1. Upon completion of either vibration; or both, the sight shall be examined to determine that no evidence of damage exists.

4.6.6.1 Vibration A.- Perform this test in compliance with 3.4.4, 3.4.4.1, and 4.6.6 and in accordance with Procedure I (Parts 1 and 2) of MIL-STD-810. Adjust the light intensity controls for maximum lamp illumination and throughout the first hour of each three hour vibration in each plane, record all predominant frequencies exhibited by the sight. Upon completion of each first hour of vibration, select the four most predominant resonant frequencies from among those recorded in each plane to acquire a maximum of twelve. During the remaining two hour vibration in each plane, vibrate the sight for at least 30 minutes at each of the four resonant frequencies previously selected. This shall result in twelve 30 minute periods - four in each plane - for a total of six hours - in addition to three first hour vibration periods, which altogether shall constitute a nine hour vibration period. The recorded line of sight position obtained and recorded in test 4.6.7.6 shall again be examined upon completion of this test - to determine that it has not changed more than one mil.

4.6.6.2 Vibration B test.- Adjust the light intensity control for maximum illumination of instrument lamps, and adjust the test rheostat (4.5.2.3c) for optimum viewing of the projected reticle image and conduct this test under conditions specified in Phase I and Phase II of 3.4.4.2. Utilizing a test target - so located that it is not vibrated during this test - establish coincidence between the central dot of the projected reticle image and the geometric center of the target circle. Throughout the entire vibration period - during both phases of vibration B - examine the established coincidence to determine that no change occurs greater than two mils in any direction.

4.6.7 General performance and optical performance test requirements.- Tests herein shall be performed under the standard ambient test conditions in the "Test Facilities" requirement of MIL-F-13926, utilizing test equipment conforming to 4.5 and subsequent to orientating the sight as specified in 3.6.

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4.6.7.1 Inclinometer accuracy.- The inclinometer ball shall be examined for compliance with 3.5.1 before and after conducting vibration tests. Subsequent to vibration tests, the inclinometer housing may be subjected to a mechanical shock - such as tapping with the head of a six inch screw driver - to facilitate compliance with 3.5.1.

4.6.7.2 Manual operability.- Within the scope of their mechanical function, manually manipulate lamp housing assemblies to determine that they are functional. Manually move the polarizing filter lever to actuate the polarizing filter from its stowed to operating to stowed positions. Determine ease of manual operation with no binding throughout each excursion, proper spring clip function, and secure positioning of the polarizing filter in its stowed and operating position. Controls shall be manually rotated throughout their entire rotational range - from stop to stop - in a cw and ccw direction to determine that they operate smoothly with no evidence of binding or slip. Also observe during this examination that all knobs are secure to their shafts and that compatibility exists between the controls intended function and associated markings.

4.6.7.3 Torque.- This test shall be performed by applying a force normal to the Elevation/Depression Control shaft. After breakaway, the knob shall be rotated CW and CCW through its entire excursion and the torque required to maintain rotation noted. These torque values shall be within the ranges specified in 3.5.3.

4.6.7.4 Illumination.- Apply 30 vdc to the sight and rotate the light intensity control knob from its extreme ccw position of minimum lamp illumination to the extreme cw position of maximum lamp illumination. Determine that a progressive increase in lamp emitted light occurs with cw rotation of the control and a progressive decrease with ccw rotation. With the control in an extreme cw position and the test circuit switch in an "OFF" position, it shall be observed in a darkened area (having not more than 0.01 foot candle of illumination) that no light is emitted from any sight lamp. The test circuit switch shall be now sequentially placed in alternate "ON" positions and while rotating the test rheostat the performance of each filament of the dual filament reticle lamp shall be verified.

4.6.7.5 Projected reticle pattern definition.- In a darkened area - as specified in 4.6.7.4 - adjust the test rheostat to obtain optimum viewing of the projected reticle pattern image, and conduct an examination to determine that its pattern is without breaks and clearly defined. After a lapse of 15 minutes - a light adaption period - employ a uniformly illuminated field of view of at least 2300 foot lamberts. The test rheostat shall now be adjusted to determine that a position exists where the prime reticle pattern image is clearly visible (see 6.4).

4.6.7.6 Optical line of sight.- This test shall employ a medium to hold and position the sight in compliance with 3.6. Prior to inserting the sight into the medium provided, utilize an auxiliary telescope whose line of sight has been orientated in accordance with 3.6.1 and whose optical axis passes through the geometric center of the beamsplitter subsequent to sight insertion. After mounting the sight, observe that its line of sight is in compliance with 3.6.1 and record the results of this examination. Subsequent to vibration this examination shall be again repeated to determine compliance with the

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line of sight retention requirement of Vibration (see 3.4.4).

4.6.7.7 Image tilt.- Utilize a test set-up similar to 4.6.7.6. Use an auxiliary telescope with a field of view (FOV) greater than 100 mils having a reticle whose vertical line has been plumbed. Insert the sight in the medium provided, apply the required electrical potential, adjust the test rheostat control for optimum viewing of the projected reticle image, and conduct the following examination. With one extremity of the projected reticle pattern image in coincidence with the end of the telescope reticle line, the other extremity - 100 mils above - shall not deviate in azimuth more than 1.7 mils to conform with 3.6.2.

4.6.7.8 Elevation and depression accuracy.- A test set-up similar to 4.6.7.6 and 4.6.7.7 shall be employed, and the line of sight in 4.6.7.6 shall be regarded as zero elevation. Conduct this examination by rotating the elevation control to elevate the sight's line of sight 125 mils above its zero elevation position as read on the auxiliary telescope reticle. Determine compliance with 3.6.3. Remove the backlash, reset the sight's line of sight to the zero elevation position in 4.6.7.6, and then utilize the elevation control to depress the sight's line of sight 125 mils. Determine compliance with 3.6.3.

4.6.7.9 Plumb travel.- This test may be conducted in conjunction with 4.6.7.8. A deviation of one mil on either side of a plumbed vertical line of the telescope reticle throughout an excursion of 100 mils above and below zero elevation shall constitute the maximum error permitted during this examination.

4.6.7.10 Reticle pattern accuracy.- View the projected reticle pattern image through an auxiliary telescope and establish coincidence between one extremity of its 100 mil horizontal line and the corresponding end of the 100 mil horizontal line of the projected reticle pattern image. Examine the azimuth extremities of both 100 mil horizontal lines at their other end and observe any lack of coincidence between them. Any existing error shall be no greater than that permitted in 3.6.5. Utilize a similar procedure using the 100 mil vertical lines of the auxiliary telescope and projected reticle pattern image to determine compliance with 3.6.5.

4.6.7.11 Parallax.- Utilize an aperture plate having at least 1.75 inch aperture concentric to the line of sight and in the same plane as the observers eye. Adjust the test rheostat for optimum viewing of the projected reticle pattern image and establish the coincidence specified in 4.6.7.10. The observer shall place his eye 8 to 10 inches from the beam-splitter and determine compliance with 3.6.6 while moving his head upward, downward, toward the left and toward the right.

4.6.7.12 Cleanliness.- Subsequent to conducting all tests and examinations the optical surfaces of the sight shall be examined in accordance with MIL-O-13830 to determine compliance with 3.3.1(c).

4.6.8 Interchangeability.

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4.6.8.1 Components.- Subsequent to meeting the requirements of 3.3, 3.4 3.5 and 3.6 in accordance with 4.1, 4.3 and 4.4, perform this test in accordance with 4.2 and 4.4.2 by disassembling and reassembling each head and body assembly as follows:

a. Divide head and body assembly components into three groups of nominated components (to obtain six groups - three groups for each disassembled assembly) as depicted in Table V-1 for the head and Table VI-1 for the body.

b. Distribute the groups of Table V-1 into three separate areas as shown in Table V-2 and distribute the groups of Table VI-1 into three separate areas as shown in Table VI-2. Each area of Table V-2 shall contain components for a complete head assembly and each area of Table VI-2 shall contain components for a complete body assembly.

c. Take in order, groups of components for the first head assembly and place in areas A, B, and C, respectively, of Table V-2, and likewise: the second in B, C, and A and the third in C, A, and B.

d. Take in order, groups of components for the first body assembly and place in areas A, B, and C, respectively, of Table VI-2, and likewise: the second in B, C, and A, and the third in C, A, and B.

*e. Reassemble each head and body assembly using only components which are in the same area. Verify that each reassembled head assembly and body assembly will reassemble into one complete sight and also verify that each reassembled sight will meet the performance requirements of 3.5 and 3.6 in accordance with 4.3.

TABLE V-1: GROUPS OF HEAD ASSEMBLY NOMINATED COMPONENTS

(GROUP 1)	(GROUP 2)	(GROUP 3)
10553401	10553274	10553422
10553298	10553390	10553300
10553268	10553269	10553329
10553297	10553327	10553328
	Assembly 10553265 less components listed in Groups 1,2, and 3.	

TABLE V-2: COMPONENT DISBURSEMENT - AFTER INTERCHANGE

	(AREA "A")	(AREA "B")	(AREA "C")
Head assembly No. 1	Group 1	Group 2	Group 3
Head assembly No. 2	Group 2	Group 3	Group 1
Head assembly No. 3	Group 3	Group 1	Group 2

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TABLE VI-1: GROUPS OF BODY ASSEMBLY NOMINATED COMPONENTS

(GROUP 1)	(GROUP 2)	(GROUP 3)
10553254	MS16105-2	10553402
10553335	10553334	MS51021-31
MS20813-1	MS51377-2	8643632

Assembly 10553258 less components listed in Groups 1, 2, and 3.

TABLE VI-2: COMPONENT DISBURSEMENT - AFTER INTERCHANGE

	(AREA "A")	(AREA "B")	(AREA "C")
Body assembly No. 1	Group 1	Group 2	Group 3
Body assembly No. 2	Group 2	Group 3	Group 1
Body assembly No. 3	Group 3	Group 1	Group 2

*Commercial components such as seals, sealants, gaskets, cements, packing, screws, nuts, washers, or pins which are rendered unserviceable by disassembly shall be replaced upon reassembly by original type equipment without penalty to this test.

"CAUTION: Head assemblies and body assemblies shall be disassembled only to the extent required to segregate nominated components in accordance with standard maintenance and repair procedures."

4.6.8.2 Assembly.- Perform this test after meeting the requirements of Section 3 in accordance with all other tests except 4.6.7.12 and 4.6.7.13.1. Randomly select (see 4.4.1) completely assembled sights and mark the head assembly and body assembly of the first sight selected as number one. Each successive selection shall be marked with the next higher number for future identification. Disassemble selected sights. Reassemble head and body assemblies which do not have identical numbers. Verify that completely reassembled sights meet all requirements of 3.5, 3.6, and 3.7.2.

4.6.9 Reliability demonstration test.- Verify compliance with 3.8 by subjecting at least three (3) sights to a minimum of 45 test cycles - each sight to at least the 15 test cycles depicted in Table VII without a failure * ; determining compliance with test related requirements - as specified under the "Test Method" column of Table VII, and by use of the following relationship:

$$(1-C) \geq \binom{N}{r} R^{(N-r)} \cdot (1-R)^{r}$$ where:

C= Confidence level
N= Total number of test cycles
R= Reliability
r= Number of failures

$$\binom{N}{r} = \frac{N!}{(N-r)!\ r!}$$

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TABLES VII - TEST CYCLES

TEST CYCLE#	EXAMINATION & TEST CYCLE	EXAMINE & TEST PER: (Perf Req't)	(Test Req't)	TEST METHOD
1. -	Storage at -65°F	3.4.1.	4.6.1	After each test cycle verify. Compliance with a Gen. & Optical Performance Requi ments of 3.5 and
2. -	Storage at +160°F	3.4.1	4.6.1	
3. -	Humidity	3.4.3	4.6.4	
4. -	Vib "A" at +60° to +90°F	3.4.4; 3.4.4.1	4.6.6; 4.6.6.1	
5. -	Vib "B" at +60° to +90°F (PHASE I)	3.4.4; 3.4.4.2a	4.6.6; 4.6.6.2	
6. -	Vib "B" at +60° to +90°F (PHASE II)	3.4.4; 3.4.4.2b	4.6.6; 4.6.6.2	
7. -	Operability at -40°F	3.4.2	4.6.2	During each test cycle verify comp ance with require 3.4.2.
8. -	Operability at +125°F	3.4.2	4.6.2	
9. -	General & Optical Perf at +60° to +90°F	3.5; 3.5.1 to 3.5.7; 3.6; 3.6.1 to 3.6.6	4.6.3; 4.6.5; 4.6.7; 4.6.7.1 to 4.6.7.12	During each test cycle verify comp ance with all Gen & Optical Perf. Requirements of 3 and 3.6.
10. -	" " "			
11. -	" " "			
12. -	" " "			
13. -	" " "			
14. -	" " "			
15. -	" " "			

* Sights which fail during an environmental exposure shall be repaired and re-tested at that environment.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with the applicable Packaging Data Sheet specified in 2.1 The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- When mounted with the M28 Armament Subsystem, the sight is primarily used to aim the wing store armament (rockets and/or 7.62mm gun) from the pilots position of the AH-1G Helicopter. The sight is fixed

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in azimuth by an initial boresight alignment and provided with an elevation control knob for the introduction of ballistic corrections. The sight's electrical power requirements are supplied by the aircraft's electrical system.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Selection of applicable levels of preservation, packaging and packing
(c) Applicable packaging data sheet number (see 5.1)
(d) Applicable stock number
(e) Number and disposition of samples for first article (initial production) sample evaluation.

6.3 Cross reference of prior sight classifications.- Sights previously covered by this specification were of two types and both types were designated as being with equipment. This equipment consisted of a range potentiometer assembly which - while not an integral part of either type sight - was included with each type for logistic purposes. The type sights previously covered by this specification were as follows:

(a) Sight, Reflex: XM73 (with equipment).- This sight consisted of Body Assembly 10553357 (N9 Modified) or Body Assembly 10553258 and Head Assembly 10553358 or Head Assembly 10553265.

(b) Sight, Reflex: XM73E1 (with equipment).- This sight consisted of Body Assembly 10553258 and Head Assembly 10553265.

6.4 Ghost images.- Multiple reflections are unavoidably generated between the two surfaces of the beamsplitter and these reflections form (secondary) images of the reticle. These images cannot be eliminated but their intensity can be minimized by adjusting the reticle lamp light intensity control (see 4.5.2.4, item 5). The projected reticle image specified as the prime image in 4.6.7.5 will be the brightest image in reference to all other visible images and it will be utilized when meeting all requirements of this specification wherein image examination is required.

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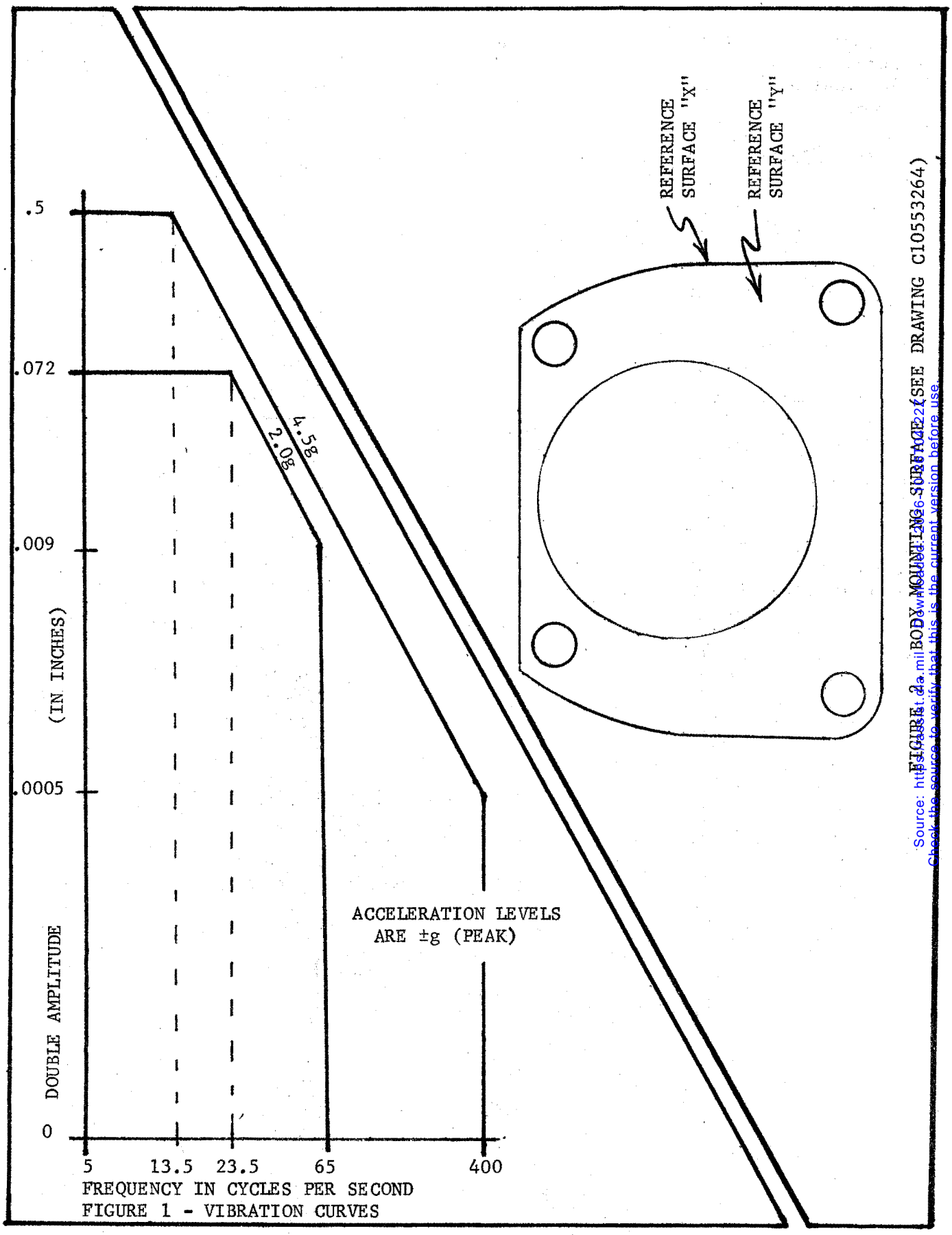


FIGURE 1 - VIBRATION CURVES

FIGURE 2 - BODY MOUNTING SURFACE (SEE DRAWING C10553264)

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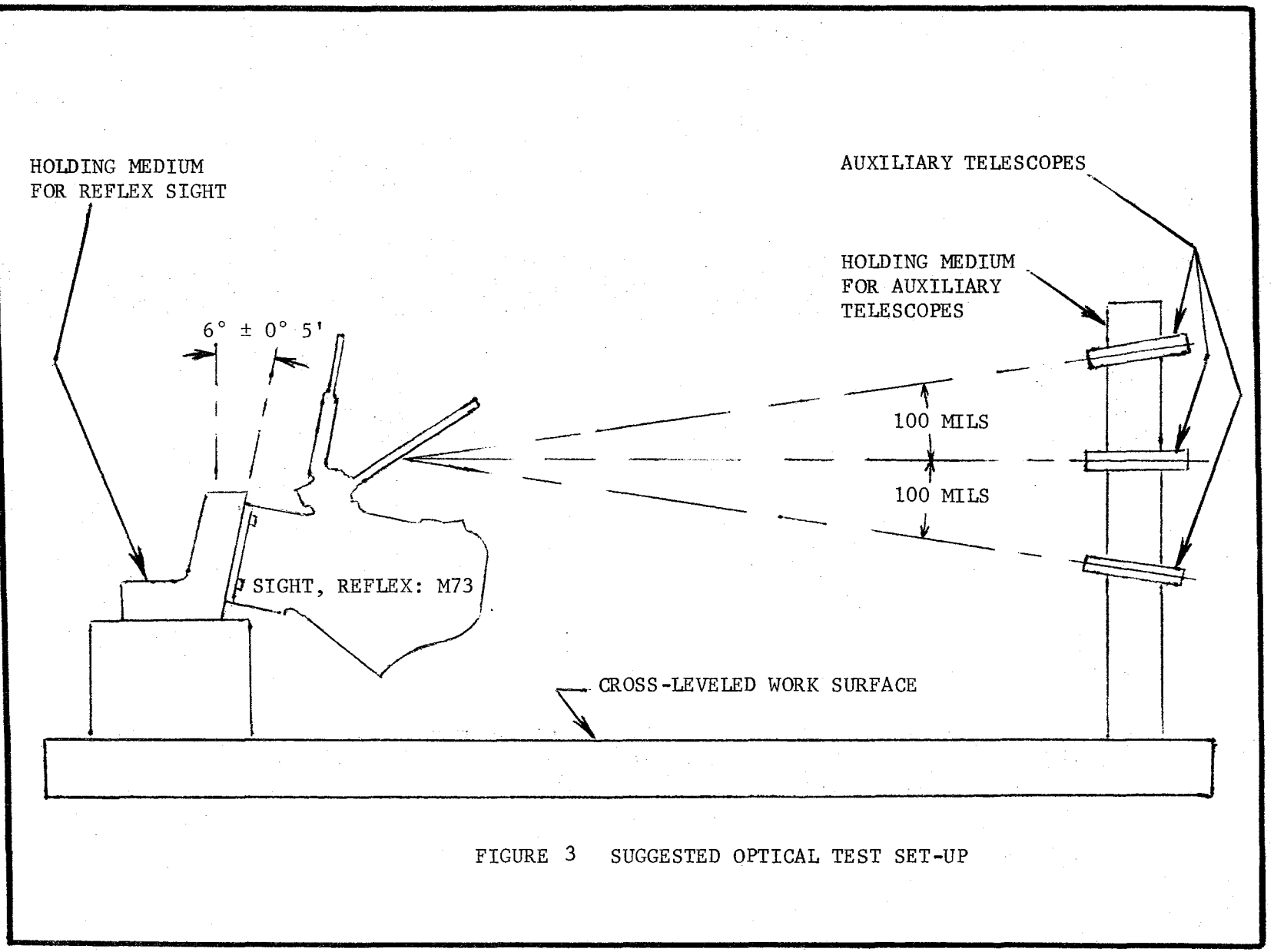


FIGURE 3 SUGGESTED OPTICAL TEST SET-UP

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